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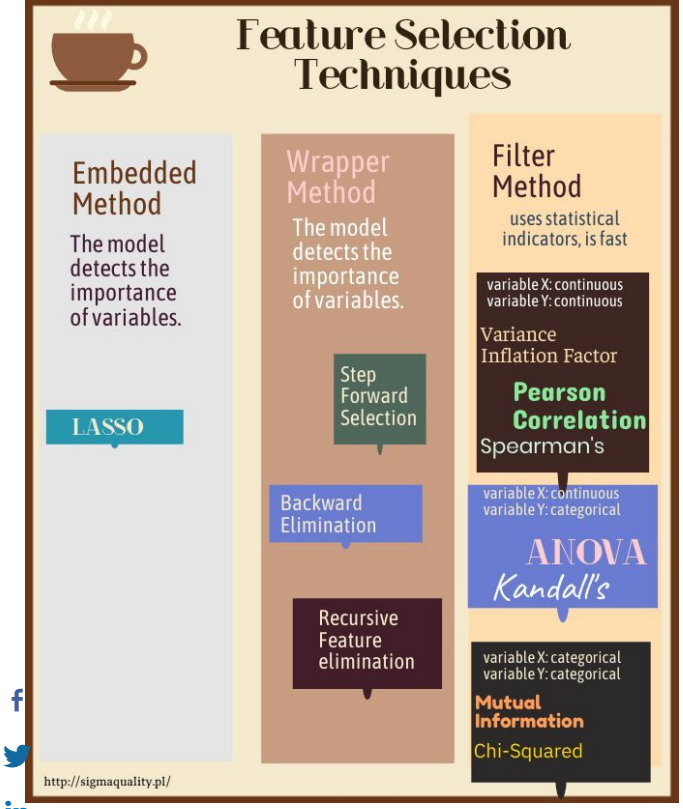
WOJCIECH MOSZCZYŃSKI

Feature Selection Techniques – Variance Inflation Factor (VIF)

March 29, 2020 admin Feature Selection Techniques 0

290320202006



Collinearity is the state where two variables are highly correlated and contain similar information about the variance within a given dataset.

The Variance Inflation Factor (VIF) technique from the Feature Selection Techniques collection is not intended to improve the quality of the model, but to remove the autocorrelation of independent variables.

In [1]:

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder,
OneHotEncoder
import warnings
warnings.filterwarnings("ignore")
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import confusion_matrix
np.random.seed(123)
```

In [2]:

Pytorch regression 2.3 [AirQualityUCI.csv]
May 2, 2020

Pytorch regression _2.1_ [WorldHappinessReport.csv]
April 30, 2020

Pytorch regression _1.1_[WorldHappinessReport]
April 29, 2020

Review of models based on gradient falling: XGBoost, LightGBM, CatBoost
April 24, 2020

Kilka prostych przykładów z programowanie objektowe w Python
April 24, 2020

Perfect Plots Bubble Plot
April 24, 2020

Perfect Plots_ Matrix of corelation
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Perfect plot Joyplot
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Homemade loop to search for the best functions for the regression model (Feature Selection Techniques)
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How to calculate the probability of survival of the Titanic catastrophe_080420201050
April 9, 2020

(B29) CatBoostClassifier #2
April 8, 2020

CatBoost Step 1. CatBoostClassifier (cat_features)
April 3, 2020

Feature Selection Techniques [categorical result] – Step Forward Selection
April 1, 2020

```
##  colorful prints
def black(text):
     print('33[30m', text, '33[0m', sep='')
def red(text):
     print('33[31m', text, '33[0m', sep='')
def green(text):
     print('33[32m', text, '33[0m', sep='')
def yellow(text):
     print('33[33m', text, '33[0m', sep='')
def blue(text):
     print('33[34m', text, '33[0m', sep='')
def magenta(text):
     print('33[35m', text, '33[0m', sep='')
def cyan(text):
     print('33[36m', text, '33[0m', sep='')
def gray(text):
     print('33[90m', text, '33[0m', sep='')
```

data source: https://www.kaggle.com/uciml/breast-cancer-wisconsin-data

In [3]:

```
df = pd.read_csv
('/home/wojciech/Pulpit/6/Breast_Cancer_Wisconsin.csv')
green(df.shape)
df.head(3)
```

```
(569, 33)
```

Out[3]:

	id	diagnosis	radius_mean	texture_mean	perimete
0	842302	M	17.99	10.38	122.8
1	842517	M	20.57	17.77	132.9
2	84300903	M	19.69	21.25	130.0

3 rows × 33 columns

Deleting unneeded columns¶

In [4]:

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```
df['concave_points_worst'] = df['concave points_worst']
df['concave_points_se'] = df['concave points_se']
df['concave_points_mean'] = df['concave points_mean']

del df['Unnamed: 32']
del df['diagnosis']
del df['id']
```

In [5]:

```
df.isnull().sum()
```

Out[5]:

```
radius_mean               0
texture_mean              0
perimeter_mean            0
area_mean                 0
smoothness_mean           0
compactness_mean          0
concavity_mean            0
concave points_mean       0
symmetry_mean             0
fractal_dimension_mean    0
radius_se                 0
texture_se                0
perimeter_se              0
area_se                   0
smoothness_se             0
compactness_se            0
concavity_se              0
concave points_se         0
symmetry_se               0
fractal_dimension_se      0
radius_worst              0
texture_worst             0
perimeter_worst           0
area_worst                0
smoothness_worst          0
compactness_worst         0
concavity_worst           0
concave points_worst      0
symmetry_worst            0
fractal_dimension_worst   0
concave_points_worst      0
concave_points_se         0
concave_points_mean       0
dtype: int64
```

In [6]:

September 2019

May 2019

March 2019

September 2018

September 2017

May 2017

March 2017

February 2017

May 2015

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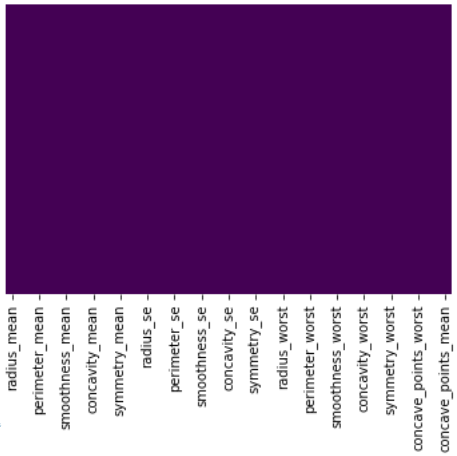
```
import seaborn as sns

sns.heatmap(df.isnull(),yticklabels=False,cbar=False,cmap=
'viridis')
```

Out[6]:

```
<matplotlib.axes._subplots.AxesSubplot at 0x7fbfd5915f10>
```



Deletes duplicates¶

there were no duplicates

In [7]:

```
green(df.shape)
df.drop_duplicates(keep='first', inplace=True)
blue(df.shape)
```

```
(569, 33)
(569, 33)
```

In [8]:

```
blue(df.dtypes)
```

```
radius_mean                float64
texture_mean               float64
perimeter_mean             float64
area_mean                  float64
smoothness_mean            float64
compactness_mean           float64
concavity_mean             float64
concave points_mean        float64
symmetry_mean              float64
fractal_dimension_mean     float64
radius_se                  float64
texture_se                 float64
perimeter_se               float64
area_se                    float64
smoothness_se              float64
compactness_se             float64
concavity_se               float64
concave points_se          float64
symmetry_se                float64
fractal_dimension_se       float64
radius_worst               float64
texture_worst              float64
perimeter_worst            float64
area_worst                 float64
smoothness_worst           float64
compactness_worst          float64
concavity_worst            float64
concave points_worst       float64
symmetry_worst             float64
fractal_dimension_worst    float64
concave_points_worst       float64
concave_points_se          float64
concave_points_mean        float64
dtype: object
```

In [9]:

```
df.columns
```

Out[9]:

```
Index(['radius_mean', 'texture_mean', 'perimeter_mean',
'area_mean',
       'smoothness_mean', 'compactness_mean',
'concavity_mean',
       'concave points_mean', 'symmetry_mean',
'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se',
'area_se', 'smoothness_se',
       'compactness_se', 'concavity_se', 'concave
points_se', 'symmetry_se',
       'fractal_dimension_se', 'radius_worst',
'texture_worst',
       'perimeter_worst', 'area_worst',
'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave
points_worst',
       'symmetry_worst', 'fractal_dimension_worst',
'concave_points_worst',
       'concave_points_se', 'concave_points_mean'],
      dtype='object')
```

We choose the continuous variable – compactness_mean¶

In [10]:

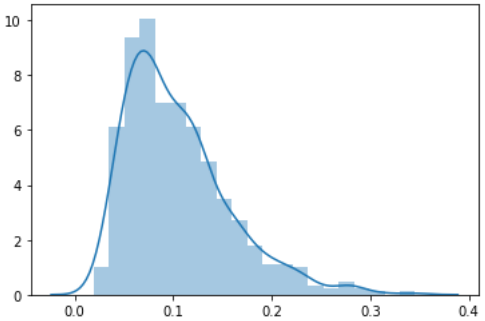
```
print('max:',df['compactness_mean'].max())
print('min:',df['compactness_mean'].min())

sns.distplot(np.array(df['compactness_mean']))
```

```
max: 0.3454
min: 0.01938
```

Out[10]:

```
<matplotlib.axes._subplots.AxesSubplot at 0x7fbfd42f2610>
```



Variance Inflation Factor (VIF)¶

In [11]:

```
import pandas as pd
import statsmodels.formula.api as smf

def get_vif(exogs, data):
    '''Return VIF (variance inflation factor) DataFrame

    Args:
    exogs (list): list of exogenous/independent variables
    data (DataFrame): the df storing all variables

    Returns:
    VIF and Tolerance DataFrame for each exogenous
variable

    Notes:
    Assume we have a list of exogenous variable [X1, X2,
X3, X4].
    To calculate the VIF and Tolerance for each variable,
we regress
    each of them against other exogenous variables. For
instance, the
    regression model for X3 is defined as:
                        X3 ~ X1 + X2 + X4
    And then we extract the R-squared from the model to
calculate:
                    VIF = 1 / (1 - R-squared)
                    Tolerance = 1 - R-squared
    The cutoff to detect multicollinearity:
                    VIF > 10 or Tolerance < 0.1
    '''

    # initialize dictionaries
    vif_dict, tolerance_dict = {}, {}

    # create formula for each exogenous variable
    for exog in exogs:
        not_exog = [i for i in exogs if i != exog]
        formula = f"{exog} ~ {' + '.join(not_exog)}"

        # extract r-squared from the fit
        r_squared = smf.ols(formula,
data=data).fit().rsquared

        # calculate VIF
        vif = 1/(1 - r_squared)
        vif_dict[exog] = vif

        # calculate tolerance
        tolerance = 1 - r_squared
        tolerance_dict[exog] = tolerance

    # return VIF DataFrame
```

```
    df_vif = pd.DataFrame({'VIF': vif_dict, 'Tolerance':
tolerance_dict})

    return df_vif
```

In [12]:

```
# import warnings
# warnings.simplefilter(action='ignore',
category=FutureWarning)
import pandas as pd
from sklearn.linear_model import LinearRegression

def sklearn_vif(exogs, data):

    # initialize dictionaries
    vif_dict, tolerance_dict = {}, {}

    # form input data for each exogenous variable
    for exog in exogs:
        not_exog = [i for i in exogs if i != exog]
        X, y = data[not_exog], data[exog]

        # extract r-squared from the fit
        r_squared = LinearRegression().fit(X, y).score(X,
y)

        # calculate VIF
        vif = 1/(1 - r_squared)
        vif_dict[exog] = vif

        # calculate tolerance
        tolerance = 1 - r_squared
        tolerance_dict[exog] = tolerance

    # return VIF DataFrame
    df_vif = pd.DataFrame({'VIF': vif_dict, 'Tolerance':
tolerance_dict})

    return df_vif
```

In [13]:

```
df.columns
exogs =['radius_mean', 'texture_mean', 'perimeter_mean',
'area_mean',
       'smoothness_mean', 'compactness_mean',
'concavity_mean',
       'concave_points_mean', 'symmetry_mean',
'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se',
'area_se', 'smoothness_se',
       'compactness_se', 'concavity_se',
'concave_points_se', 'symmetry_se',
       'fractal_dimension_se', 'radius_worst',
'texture_worst',
       'perimeter_worst', 'area_worst',
'smoothness_worst',
       'compactness_worst', 'concavity_worst',
'concave_points_worst',
       'symmetry_worst', 'fractal_dimension_worst']
```

In [14]:

```
print('Jeżeli VIF wynosi więcej niż 5 prawdopodobnie
występuje multicollinearity' )

pks = sklearn_vif(exogs, df)
pks.sort_values('VIF').round(1)
print()
blue('LinearRegression in sklearn')
blue(pks[pks['VIF']<=10])


kot = get_vif(exogs, df)
kot.sort_values('VIF').round(1)
print()
green('LinearRegression in statasmodels')
green(kot[kot['VIF']<=10])
```

```
Jeżeli VIF wynosi więcej niż 5 prawdopodobnie występuje
multicollinearity

LinearRegression in sklearn
                          VIF  Tolerance
smoothness_mean       8.194282   0.122036
symmetry_mean         4.220656   0.236930
texture_se            4.205423   0.237788
smoothness_se         4.027923   0.248267
symmetry_se           5.175426   0.193221
fractal_dimension_se  9.717987   0.102902
symmetry_worst        9.520570   0.105036

LinearRegression in statasmodels
                          VIF  Tolerance
smoothness_mean       8.194282   0.122036
symmetry_mean         4.220656   0.236930
texture_se            4.205423   0.237788
smoothness_se         4.027923   0.248267
symmetry_se           5.175426   0.193221
fractal_dimension_se  9.717987   0.102902
symmetry_worst        9.520570   0.105036
```

OLS linear regression model for variables before reduction¶

In [15]:

```
blue(df.shape)
```

```
(569, 33)
```

In [16]:

```
X1 = df.drop('compactness_mean', axis=1)
y1 = df['compactness_mean']
```

In [17]:

```
from statsmodels.formula.api import ols
import statsmodels.api as sm

model = sm.OLS(y1, sm.add_constant(X1))
model_fit = model.fit()

print('R2: %.6f' % model_fit.rsquared)
#blue(model_fit.summary())
```

```
R2: 0.980200
```

OLS linear regression model for variables after reduction¶

In [22]:

```
df2
=df[['smoothness_mean','symmetry_mean','texture_se','smoot
hness_se',
'fractal_dimension_se','symmetry_worst','compactness_mean'
]]
```

In [23]:

```
X2 = df2.drop('compactness_mean', axis=1)
y2 = df2['compactness_mean']
```

In [24]:

```
from statsmodels.formula.api import ols
import statsmodels.api as sm

model = sm.OLS(y2, sm.add_constant(X2))
model_fit = model.fit()

print('R2: %.6f' % model_fit.rsquared)
#blue(model_fit.summary())
red('The reduction of dimensions caused the deterioration
of the models properties')
```

```
R2: 0.649990
The reduction of dimensions caused the deterioration of
the models properties
```

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Feature Selection Techniques [numerical result] – Step Forward Selection